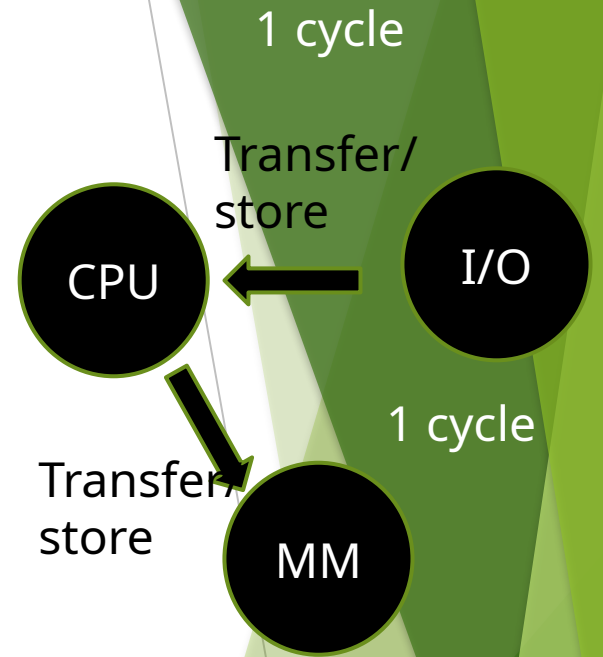


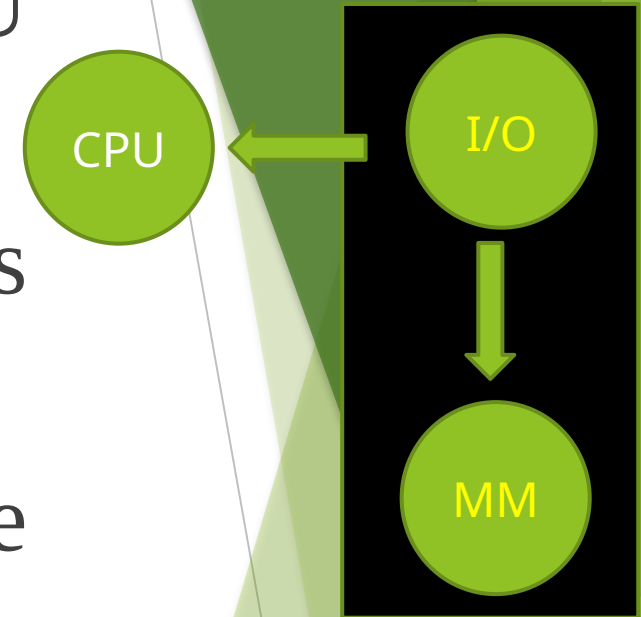
The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect. The shapes are concentrated on the left and right sides, leaving a large white central area.

DMA

- ▶ Why DMA?
- ▶ Basically we have the CPU, I/O, and memory
- ▶ Suppose the I/O devices want to transfer data to CPU, then CPU will transfer it to the memory. Whenever we have to access memory, first I/O goes to CPU and CPU access memory. For these, 2 cycles are required to transfer from I/O to memory and it's time-consuming. Here the data transfer rate is slow.
- ▶ This is the normal procedure without the concept of DMA.

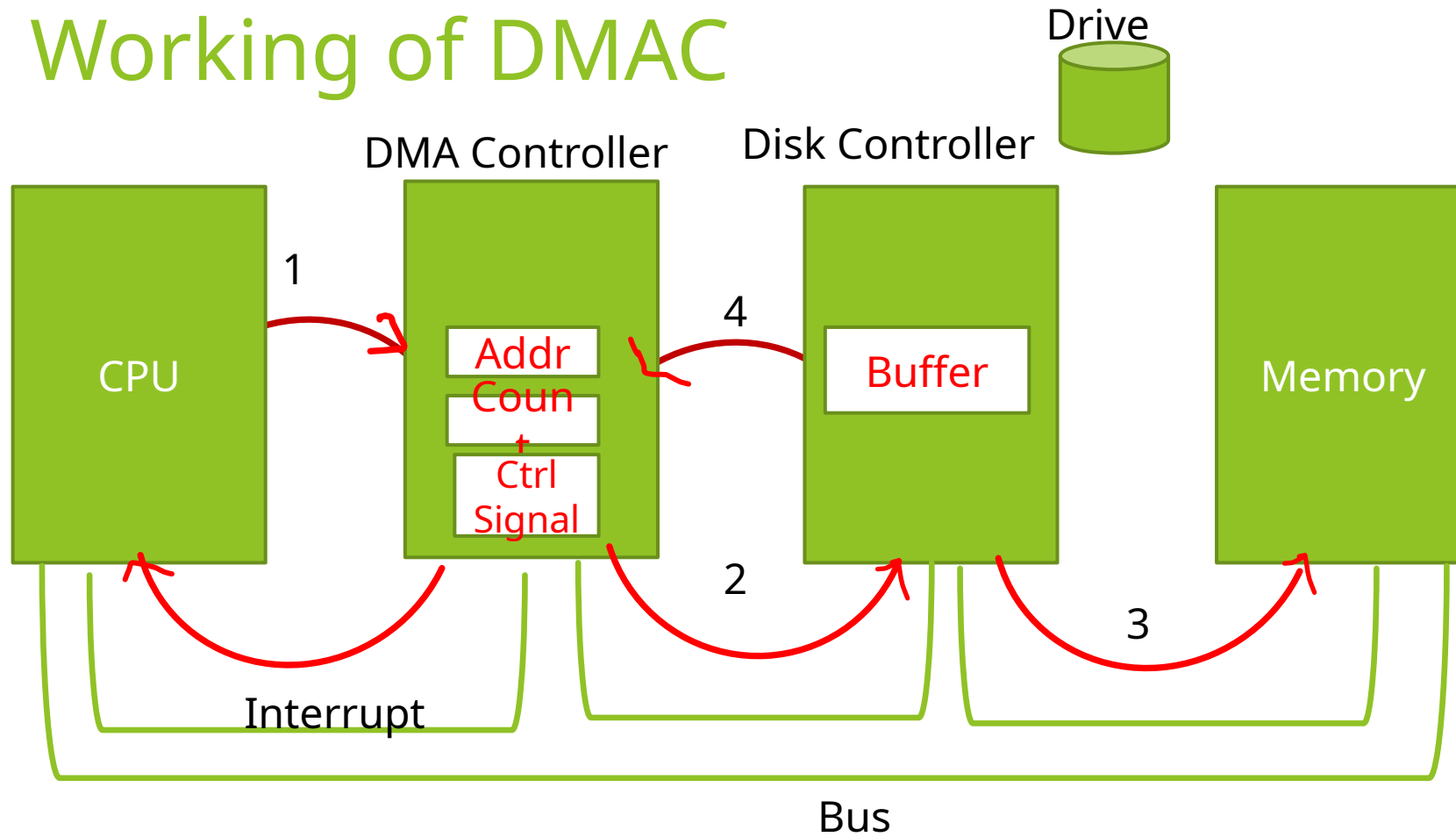


- ▶ Suppose if there is a direct connection from I/O to memory without the CPU involvement.
- ▶ So it will save one cycle, and here comes the role of DMA.
- ▶ This connection bw the I/O and memory ie DMA is provided by DMA controller



- ▶ Direct memory access (DMA) is a method that allows an input/output (I/O) device to send or receive data directly to or from the main memory, bypassing the CPU to speed up memory operations.
- ▶ It is designed by Intel to transfer data at a faster rate.
- ▶ It allows the I/O device, to transfer data directly to/from memory without the CPU involvement.
- ▶ The process is managed by a chip known as a DMA controller (DMAC).

Working of DMAC

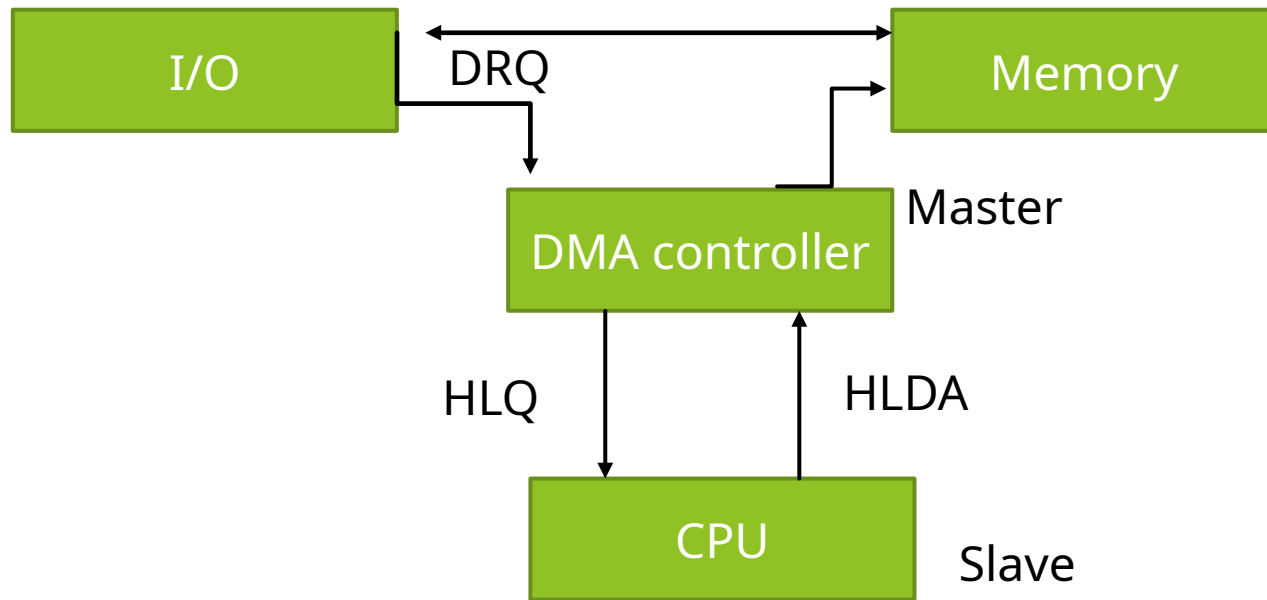


- 
- ▶ We have the following components
 - ▶ CPU
 - ▶ DMA controller
 - ▶ Disk Controller
 - ▶ Memory
 - ▶ These components are connected by Bus.

How DMA operation are performed?

- ▶ If the device wants to send data to memory, first the device has to send a DMA request (DRQ) to DMA controller.
- ▶ Then the DMA controller sends HLQ signal (Hold Request) to CPU and waits for the CPU to send ack (HLDA). It is asking you to remain in hold and hand over the system bus to me so that I can transfer the data directly from peripherals to memory.

- ▶ Whenever the DMA gets the Hold Ack signal (HLDA), it becomes the master and CPU become the slave.
- ▶ Then the CPU leaves the control over bus and acknowledge the Hold ack signal (HLDA) to DMA controller.
- ▶ Now, the CPU in hold state, and the DMA controller has to manage operations over buses between CPU, memory and I/O devices.



END